



From Scratch to Python



Lesson Objectives

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- Understand which data type a variable is.
- Understand why a computer needs to know what the data type is for a given input/variable.
- Learn how to change the data type of a variable.

Lesson Outcomes

- Carry out a number of programming tasks to practice programming simple arithmetic

Literacy – Key Words

Input	Values which get sent from the user into the computer
Variable	The place where inputs get stored by the program
Output	The values which get sent from the computer to the user



Review of last few lessons



Can you tell me the following?

1. The python code needed to display a word on the screen? 

```
1 print("hello")  
2
```

hello

2. The python code needed to ask the user for an input? 

```
input("What is your name?")
```

3. The way python can store user inputs? 

```
answer = input("What is your name?")
```



Programming Inputs & Storing Inputs in PYTHON

In python:

```
1 answer = input("What is your name?")
2 print(answer)
3
```



coder.computerscienceuk.com says

What is your name?

OK Cancel



In Python, the 'input' statement would allow our program to ask the user to enter an input.

The 'answer' variable would store the input.

And we could use the answer variable with a 'print' statement to output the inputted value.

Output



```
1 answer = input("What is your name?")
2 print(answer)
3
```

Mr Wickins



What is the Major Difference between a print and an input statement?

```
1 print("hello")  
2
```

hello

PRINT Just displays a message!

```
input("What is your name?")
```

INPUT Displays a message and waits for a user input!

```
1 answer = input("What is your name?")  
2 print(answer)  
3
```

Mr Wickins

So, it needs to equal a variable if we want to store what the user types in...



Today:

Data Types and Arithmetic



What is a data type?

- What different types of data are there?

String

This means any combination of keyboard characters (letters, numbers symbols)

Integer

This means any whole number. If the computer knows the data is an integer it can do maths with the data

Real

This means any decimal (fractional) number



The need to tell the computer what data type the variable will store

- Computers store values in variables



- In order for the computer to understand our inputs, it needs to **know** what **type of data** our input is.
- If a variable is to hold a **word**, we must tell the computer that the variable is to contain a **string**.
- If a variable is to hold a **whole number**, we must tell the computer that the variable is to contain an **integer** – so it can do calculations.



Doing Arithmetic

- If we want our program to work with numbers we must tell the program that certain variables will hold numbers.
- When we input a number into our program...

```
num1 = input("please enter a number")
```

...python automatically sets the variable as a string *(it doesn't recognise it as a number, instead it simply recognises it as a keyboard symbol)*

- If we want to do some maths with the 'inputs' we need to tell the computer that the variable contains a number (integer).



Converting Data Types

- To do this we need to convert the variable to an 'integer' data type.

```
1 num1 = input("Please enter a number: ")
2 num2 = input("Please enter another number: ")
3
4 num1 = int(num1)
5 num2 = int(num2)
```

Above you can see that two numbers have been entered and stored in variables num1 and num2.

As python stores inputs as strings we need to convert these variables into integers.

The int() function allows us to do this!

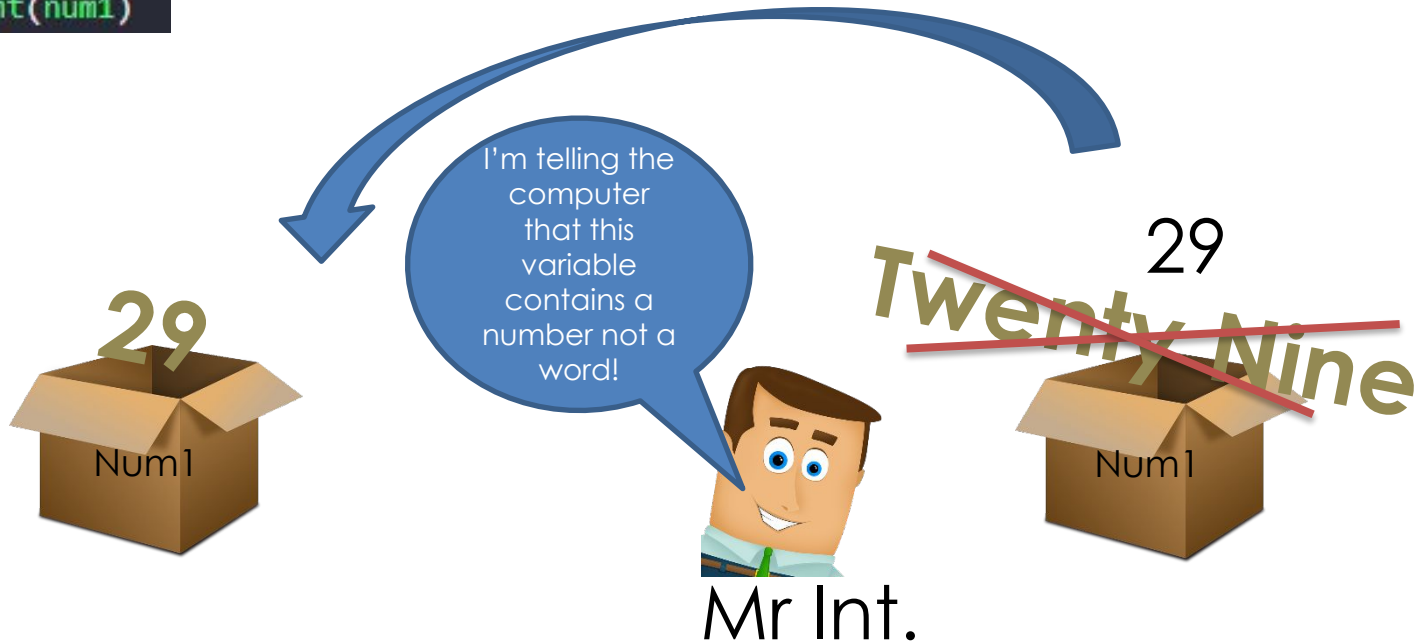


What's going on?

```
num1 = input("Please enter a number: ")
```

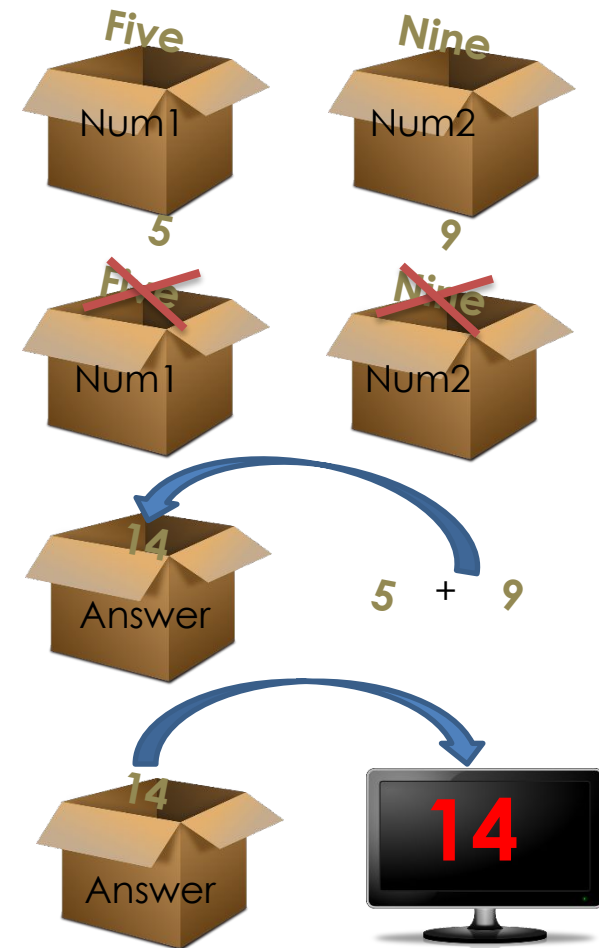


```
num1 = int(num1)
```



Once they have been converted we're ready to do maths!

```
1 num1 = input("Please enter a number: ")
2 num2 = input("Please enter another number: ")
3
4 num1 = int(num1)
5 num2 = int(num2)
6
7 answer = num1 + num2
8
9 print("Your answer is: ", answer)
```



Lesson Activity

- Read the information in the lesson's workbook
- Complete the PRIMM tasks provided.

